Adediwura Afolabi

Mechanical Design Engineer | Mechatronics Engineering Student

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PROFILE SUMMARY

Results-driven Mechatronics Engineering student and Mechanical Design Engineer with hands-on experience in 3D CAD modelling, mechanical system assembly, and production-ready drawing generation. Skilled in SOLIDWORKS modelling, motion simulation, and design-for-manufacture principles. Strong analytical thinker with disciplined documentation practices and cross-functional collaboration experience.

EDUCATION

FEDERAL UNIVERSITY OF TECHNOLOGY AKURE (FUTA) **Akure, Nigeria**

Bachelor of Engineering in Mechatronics Engineering (4.70/5.0) **June 2023 – November 2027**

TECHNICAL SKILLS

- CAD Modelling
- SOLIDWORKS Simulation
- 2D Drawing
- SOLIDWORKS
- MATLAB
- Google Tools e.g. Google Docs, Meet, Calendar, Sheets for cross documentation and collaboration
- Canva for idea presentation
- Critical Thinking
- Problem-solving, Communication skills, and Attention to detail
- Leadership skills
- Creativity and Innovation
- Arduino Uno

PROFESSIONAL EXPERIENCE

SMART SYSTEMS RESEARCH LABORATORY **Akure, Nigeria**

Mechanical Engineering Design Intern (SWEP II) **Sept. 2025 – Oct. 2025**

- Designed and modelled precision mechanical components (flanges, collars, structural fittings, mechanical tools) using SOLIDWORKS with emphasis on dimensional accuracy, and manufacturability.
- Developed a fully assembled robotic arm system, applying principles of mechanical stability for reliable operation.
- Generated detailed 2D production drawings including assembly layouts, and exploded views to support fabrication and inspection processes.

- Performed motion simulation to evaluate structural alignment and operational integrity.

BOOTCLASS **Akure, Nigeria**

Corporate Communications Intern (Undergraduate Intern) **Feb. 2023 – Jan. 2025**

- Produced structured technical documentation and organized reports to support operational traceability and process documentation.
- Maintained detailed records and version-controlled documentation for internal reference and compliance purposes.
- Coordinated cross-functional communication to ensure accurate transmission of technical and operational information.
- Developed structured reporting systems to improve clarity, accountability, and information retention.

RECENT PROJECTS

Industrial Robotic Arm : Automated Handling System | SOLIDWORKS

- Designed and assembled a multi-joint robotic arm system for automated material handling applications.
- Applied kinematic analysis and constraint evaluation to ensure mechanical stability and controlled motion.
- Generated complete manufacturing drawings including assembly layouts and exploded configurations.
- Simulated operational movement to validate range of motion and mechanical reliability.

Arduino Digital Weighing Scale | Arduino Uno, Sensor Interfacing

- Designed and built a functional digital weighing scale using Arduino Uno, HX711 load cell sensors (50kg), and supporting electronics using **Arduino programming**.
- Integrated sensors with Arduino to accurately measure weight, and displayed results digitally using sensor interfacing.
- Constructed a stable wooden platform for practical use.

Domestic Gas Burner System – Mechanical & Structural Design | SOLIDWORKS

- Modelled interacting components including burner head, gas inlet, support structures, and vessel assembly.
- Developed exploded assembly views and detailed fabrication drawings for manufacturing clarity.
- Evaluated mechanical constraints and structural stability under operational conditions.

SCHOLARSHIPS / AWARDS

- BabsFoundation Scholarship Award – One out of 60 undergraduates **April 2025**
- Most Social Volunteer of 2023/2024 Academic Session issued by BootClass **September 2024**
- Certificate of Excellence in Engineering Oratory Contest **September 2024**
- BabsFoundation Scholarship Award – One out of 40 undergraduates **July 2024**

- Best **Female** Student in School of Engineering and Engineering Technology 2022/2023 Academic session issued by FUTA Alumni North America (FUTANA)- One out of 50 undergraduates out of a competitive pool of 4500 undergraduates **July 2024**
- Certificate of Active Participation in a prototyping workshop **October 2023**

Link to all certificates: https://bit.ly/Adediwuracertificates

LEADERSHIP / EXTRACURRICULAR

- Leading a **43 students'** Mechatronics Engineering class, ensuring smooth communication between lecturers and students, fixing class schedule and ensuring lecturers conditions are met, and student grievances resolved.
- Supporting **operational efficiency** at **an international** organization through adherence to cleanliness, safety, and organizational standards, reflecting an understanding of workplace safety, 5S principles, and process discipline.
- Performed structured documentation and technical reporting at **an international** organization by accurately recording meeting minutes and maintaining attendance logs, demonstrating strong data organization, traceability, and documentation control skills applicable to engineering projects.
- Attended **prototyping and 3D printing** workshops.